

Software Requirements Specification

5.1 Introduction

The DVD Lending System is a web application for the CS department media center designed to replace a manual paper log. The primary goal is to provide real-time inventory visibility, track checkout history, and manage borrowing rules.

5.2 Functional Requirements

5.2.1 User Authentication

- As a user, I want the system to authenticate borrowers, staff, and administrators through the existing campus login system to ensure secure and easy access.

5.2.2 Inventory Search and Browse

- As a user, I want to search for DVDs by title so I can find specific media quickly.
- As a user, I want to browse the catalog by genre or availability status to see what is currently in stock.

5.2.3 DVD Checkout/Borrowing

- As a borrower, I want to check out a DVD if it is marked "available" so I can take it home.

5.2.4 DVD Return

- As a borrower or staff member, I want to return DVDs to the system so that the status updates to "available" for the next user.

5.2.5 Catalog Management

- As a staff member, I want to add new DVDs to the catalog to keep the collection up to date.
- As a staff member, I want to remove DVDs from the catalog if they are lost or retired.

5.2.6 Overdue Monitoring

- As a staff member, I want to view a list of all overdue DVDs so I can manage late returns effectively.

5.2.7 System Configuration

- As an administrator, I want to manage system configurations and generate usage reports to oversee the media center's performance.

5.3 Performance Requirements

5.3.1 Real-time Status Updates

- The system ensures data integrity by updating a DVD's availability status immediately upon checkout or return.

5.3.2 Concurrent User Access

- The application supports multiple simultaneous users browsing and searching the catalog without experiencing significant latency.

5.3.3 Authentication Response Time

- To maintain a high-quality user experience, the handshake with the campus login system completes in under 2 seconds.

5.4 Environment Requirements

5.4.1 Development Environment Requirements

- **Version Control:** A Git-based repository to facilitate iterative Agile development and CI/CD.
- **Framework:** A Next.js web stack to provide a modern, responsive user interface.
- **Database:** A Supabase relational database to track inventory status, history, and roles.

5.4.2 Execution Environment Requirements

- **Platform:** Vercel hosting to ensure the application is accessible via standard campus network protocols.
- **Client Access:** Full compatibility with modern web browsers used by students and faculty.
- **Integration:** Secure network access to the existing campus authentication API.